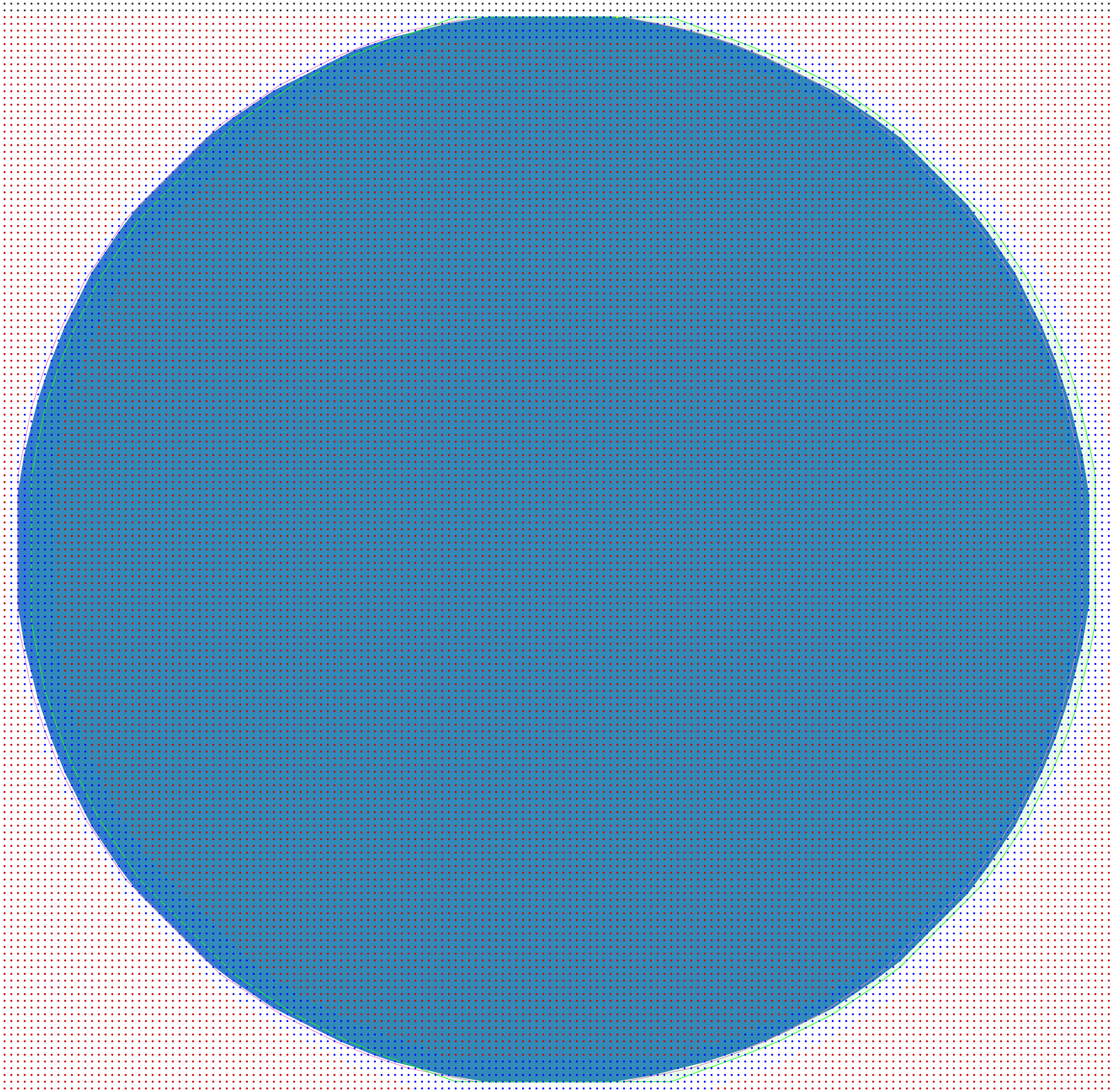

k = 19999

Circle ub on perimeter: 501.31312162815874

solution parimeter: 500.35564123690756

 # vertices: 196